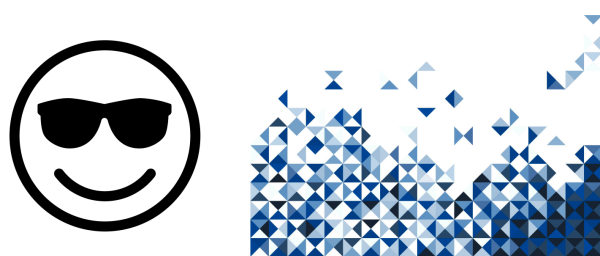


Graphical Abstract

Informational Content Analysis of Lost Languages

Richard Witt, Anamaria Berea



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Highlights

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- Bullet 1
- Bullet 2
- Bullet 3

Informational Content Analysis of Lost Languages

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Abstract

Deciphering lost scripts creates new sources of evidence to understand how language evolved, verify archaeological evidence, and rediscover lost knowledge. Low resource languages, particularly, undeciphered scripts, present a particular challenge for Natural Language Processing since the small corpus size requires us to determine if the available informational content is sufficient for decipherment to be theoretically possible. This paper outlines several methods used to aid in decipherment. Unicity distance is calculated for Rongorongo, Linear A, and the Indus Valley Script to determine decipherability. These three lost scripts are then analyzed to determine the type of writing system using Zipf's Law and HDBSCAN. By combining these two analyses it is found that the Linear A corpus does have enough informational content for decipherment regardless of assumed writing type, however the Rongorongo and Indus Valley corpora only have enough information if we assume they are logographic. [I will be able to make some statement like this, but likely not as strong.]

Keywords: Keyword1, keyword2, keyword3

TODO: Update for new article

PACS: 01.78.+p, 89.20.Dd, 01.75.+m, 89.70.Cf

TODO: Update for new article

2000 MSC: 68T50, 91B82, 62P12, 91F99

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1. Introduction

Deciphering Rongorongo, Linear A, and the Indus Valley Script would increase knowledge and understanding in archeology, anthropology, history, linguistics, astronomy and more. Still, research has been unable to yield satisfactory decipherment of these lost scripts and basic questions such as Rongorongo's status as a writing system is still an open question. Barthel created the now canonical transliterations for Rongorongo in 1958. Transliterations for Linear A and Indus Valley were provided by TODO in TODO and TODO in TODO respectively.

Current research has yielded ... TODO 1-2 sentence summary.

This paper uses these transliterations to conduct an analysis using Natural Language Processing to provide insight on two different open questions. We first determine if there is enough informational content in the corpora for theoretical decipherment to be possible. Next we provide insight on the type of writing system for each lost language.

The remainder of this paper is structured as follows: Section 2 describes initial and current efforts to decipher the lost scripts. Section 3 details the source transliterations used in the analysis. Section 4 describes the methodology and rationale used to determine decipherability and writing system. Section 5 provides the analysis and results. Finally Section 6 discusses the results and describes potential paths for future work.

2. Literature Review

2.1. Easter Island and Rongo Rongo

Efforts to translate and decipher have spanned centuries.

Rapa Nui is still the spoken language of the indigenous people of Easter Island.

Father Sebastian Englebert creates Rapa Nui to Spanish Dictionary in 1938, translated to spanish in and updated in 1978.[1]

Rapa Nui to Spanish

In his 1958 work *Grundlagen zur Entzifferung der Osterinselschrift* Barthel created the alphanumeric cataloging system and transliterations from glyph to alpha numeric characters [2].

Horley provides corrected glyphs and

Jacques Guy provided the tablet detail [3] and original web resources Philip Spaelti used to create Kohaumotu.org which provides the glyph library this paper uses as input data. [4]

2.2. TODO

3. Data

Two distinct data sets were used. One for Rongo Rongo and one for Linear A. Linear A Indus valley script is Rongo Rongo is the extinct script of Easter Island

3.1. Rongo Rongo

Both the analysis of Rongo Rongo with variants and Rongo Rongo analysis without variants was conducted using data available at the [Rongorongo corpus \(codes only\)](#). This data is [1]. Initially recorded by Barthel in 1958 [2]. Impressive interactive website greatly aiding in the understanding of Rongo Rongo glyphs. We make no assertion on the accuracy 26 Tablets Coded A though Z. from [the Kohaumotu Rongorongo codes page](#).

Create a corpus of all encodings for Rongorongo based on updated 3 character glyph codes.

Instead of using the glyphs themselves we rely directly on the index created by Barthel as it represents a manual encoding of glyphs.

Glyphs are numbered in a systematic manner. Each row is intended to categorize the glyphs with Row 0 containing the simplest forms while higher rows increase in complexity. While only 90 rows are shown 799 total rows were categorized by Barthel and available in [1] and in the Kohaumotu.org cite.

To create the index used in this analysis the row and column are concatenated to result in the three digit index used. In this manner 000 represents the blank space in the upper left. and 001 represents both variants



Figure 1: **Rongo Rongo Great Washington Tablet.** Images created by the Smithsonian and compiled by .

	0	1	2	3	4	5	6	7	8	9
0										
10										
20										
30										
40										
50										
60										
70										
80										
90										

Figure 2: Rongorongo Glyph Examples. TODO.

3.2. Linear A

4. Methodology and rationale

TODO



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Figure 3: **TODO**. TODO.

4.1. TODO

TODO

5. Analysis and Results

TODO

6. Discussion and Future Work

TODO

Appendix A. TODO

Year	Title	TODO
2001	Astronomy and Astrophysics for the New Millennium	Astrophysics
2003	The Sun to the Earth—and Beyond: A Decadal Research Strategy in Solar and Space Physics	Heliophysics
2003	New Frontiers in the Solar System: An Integrated Exploration Strategy	Planetary Science
2007	Earth Science and Applications from Space: National Imperatives for the Next Decade and Beyond	Earth Science

Table A.1: Decadal Surveys Used in Analysis

Appendix B. TODO2

This section details TODO

TODO1: Follow a `yyyy-DS-Category-####.pdf` format where `####` is the National Academies document number. `DS` is used to categorize all Decadal Surveys. `MA` replaces `DS` for Midterm Assessments, and `SP` is used for the Special Publication in 2015 that identified lessons learned.

TODO2: Follow a `yyyy_SSP_Nasa_File_Name.pdf` format. `SSP` represents Science mission directorate Science Plans. This is replaced with `AD` to indicate Additional Documentation, `NSP` to indicate NASA Science Plan, or `AD` to indicate additional documentation present on the SMD website.

References

- [1] S. Englert, Diccionario Rapanui-Español, 1948, translated from Spanish to English by Jacques Guy.
- [2] T. S. Barthel, Grundlagen zur Entzifferung der Osterinselschrift, Cram, de Gruyter, Hamburg, 1958.

- [3] J. Guy, [Rongorongo or the hieroglyphs of the easter island tablets](http://kohaumotu.org/rongorongo_org/), archived and mirrored by Philip Spaelti (2005).
URL http://kohaumotu.org/rongorongo_org/
- [4] P. Spaelti, [Glyph library](http://kohaumotu.org/Rongorongo/Glyph_Library/index.html).
URL http://kohaumotu.org/Rongorongo/Glyph_Library/index.html